

The chemical reaction between two or more substances giving rise to products is governed by certain laws. These laws are called Laws of chemical combination.

Gay Lussac's Law of Combining Volume

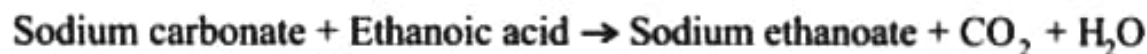
According to this law, "Mass can neither be created nor destroyed." It means During a chemical reaction total mass of reactants will be equal to total mass of products.

for example $A + B \longrightarrow AB$
(Reactant) (Product)

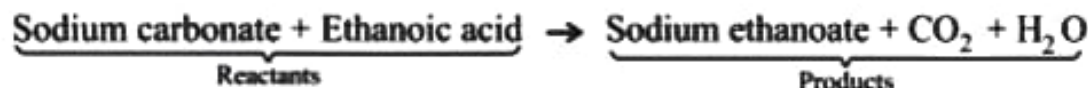
then

$$m_A + m_B = m_{AB}$$

Example : In a reaction 5.3 gm of sodium carbonate reacted with 6 gm of ethanoic acid. The products were 2.2 gm of CO_2 , 0.9 gm of H_2O and 8.2 gm of sodium ethanoate. Show that these observation are all in agreement with law of conservation of mass.



Solution :



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Now, according to the law of conservation of mass :

$$\text{Mass of sodium carbonate} + \text{Mass of ethanoic acid} = \text{Mass of sodium ethanoate} + \text{Mass of CO}_2 + \text{Mass of H}_2\text{O}$$

Putting values of masses from the equation :

$$5.3 \text{ gm} + 6.0 \text{ gm} = 8.2 \text{ gm} + 2.2 \text{ gm} + 0.9 \text{ gm}$$

$$\text{Or} \quad 11.3 \text{ gm} = 11.3 \text{ gm}$$

Since, LHS = RHS

$\therefore$ Law of conservation of mass is in agreement with the given values in equation.

DELTA & EMOTA

• Law of constant Proportion:

According to this law, "A pure chemical compound always contains the same elements combined together in the same proportion by mass irrespective of the fact from where the sample has been taken or from which procedure has it been produced."

और Example से समझो ↷

18 gm of $H_2O \Rightarrow 16$ gm of oxygen + 2 gm of hydrogen

मतलब $m_1/m_2 = 2/16 = \frac{1}{8}$ $\left\{ \begin{array}{l} \because m_1 = \text{mass of H} \\ m_2 = \text{mass of O} \end{array} \right.$

36 gm of $H_2O \Rightarrow 32$ gm of oxygen + 4 gm of hydrogen

$$m_1/m_2 = 4/32 = \frac{1}{8}$$

Compound के mass का proportion हमेशा Same रहेगा।
↖ Element के

Example : Hydrogen and oxygen combine in the ratio 1 : 8 by mass to form water. What mass of oxygen gas would be required to react completely with 3.0 gm of hydrogen gas ?

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Solution : $\frac{m_H}{m_O} = \frac{1}{8}$ Given in equation (For H_2O)

But, $m_H = 3.0 \text{ gm}$ (given)

Or $\frac{3}{m_O} = \frac{1}{8}$

Or $m_O = 24 \text{ gm}$

$\therefore$ Mass of oxygen will be 24 gm.

Or it will be a sample of 27 gm of H_2O where 3 gm of hydrogen is present with 24 gm of oxygen.

• Dalton's Atomic Theory :

- All matter is made up of very tiny particles called Atoms.
- Atoms are indivisible particles, which can't be created or destroyed in a chemical reaction. { Law of Conservation of mass }
- Atoms of an element have identical mass and chemical Properties.
- Atoms combine in the ratio of small whole numbers to form Compounds. { Law of constant proportion }
- The relative number and kinds of atoms are constant in given Compound.

■ ATOM :

- किसी Element का smallest particle Atom होता है। during the chemical reaction, the atom maintain its identity throughout chemical or physical change.
- Atoms को Eyes से नहीं देखा जा सकता, we use very very high power microscope to see Atoms.
- Atomic radius of smallest atom in hydrogen is 0.37×10^{-10} m or 0.037 nm.

देखो

$$1 \text{ nm} = 10^{-9} \text{ m}$$

nm = nanometer

IUPAC (International Union of Pure & Applied Chemistry) Symbols of Atoms of Different Elements

Element	Symbol
Aluminium	Al
Argon	Ar
Barium	Ba
Calcium	Ca
Carbon	C
Chlorine	Cl
Cobalt	Co
Copper	Cu
Fluorine	F
Gold	Au
Hydrogen	H

Element	Symbol
Iodine	I
Iron	Fe
Lead	Pb
Nitrogen	N
Oxygen	O
Potassium	K
Silicon	Si
Silver	Ag
Sulphur	S
Zinc	Zn

■ Atomic Mass :

- The mass of an atom of an element is called its atomic Mass.
- In 1961, IUPAC have accepted 'atomic mass unit' (u) to express atomic & molecular mass of elements and Compounds.

■ Atomic mass unit :

The atomic mass unit is defined as the quantity of mass equal to $1/12$ of mass of an atom of Carbon-12.

$$1 \text{ amu or } u = \frac{1}{12} \times \text{Mass of an atom of C}$$

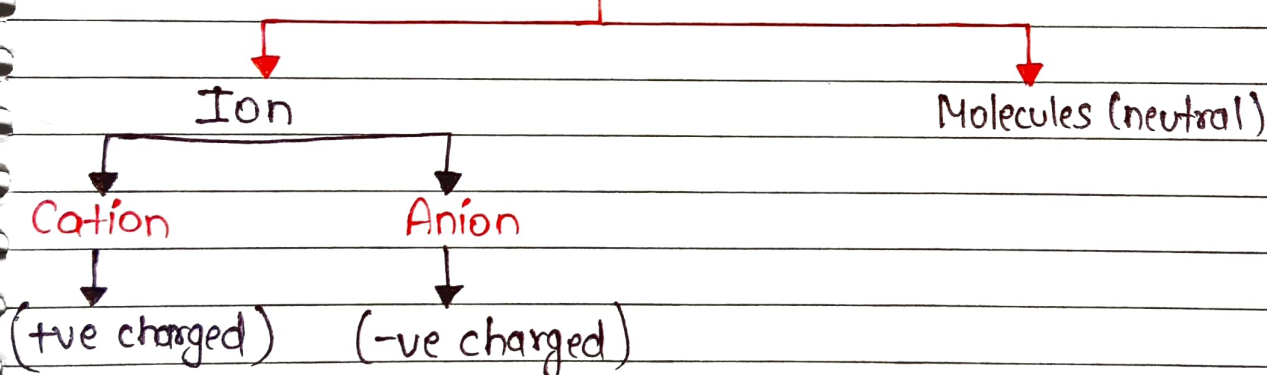
$$1u = 1.66 \times 10^{-27} \text{ Kg}$$

Element	Atomic Mass	Element	Atomic mass
Hydrogen	1u	Magnesium	24 u
Carbon	12u	Sulphur	32 u
Nitrogen	14 u	Chloride	35.5 u
Oxygen	16u	Calcium	40u
Sodium	23u		

■ How do atoms exist ?

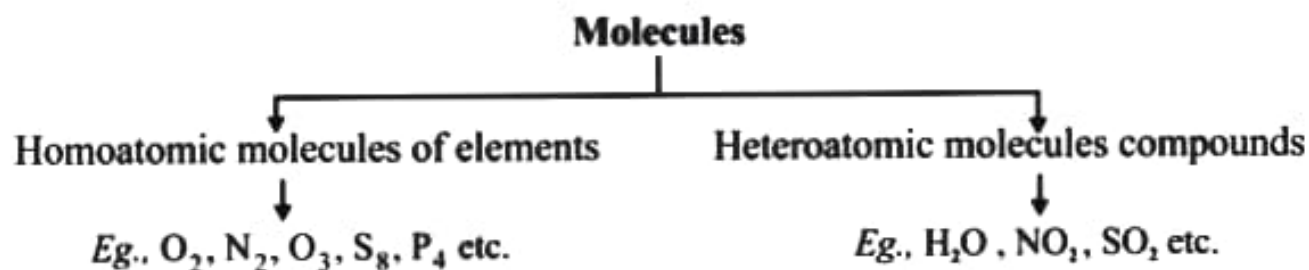
- Atoms of most of the elements are very reactive and does not exist in free state.
- Only the atoms of noble gases (such as He, Ne, Ar, Kr, Xe) are chemically unreactive and can exist in free state as single atom.
- Atoms of all other elements combine together to form molecules or ions.

Atom (Neutral)



■ Molecule 8

- A molecule is a group of two or more atoms which are chemically bonded with each other.
- A molecule is the smallest particle of matter (except element) which is capable of an independent existence and show all properties of that substance.



Atomicity

The number of atoms present in one molecule of an element is called its atomicity.

Name	Formula	Atomicity	
1. Argon	Ar	Monoatomic (1)	} Noble gasses constitute monoatomic molecules
2. Helium	He	Monoatomic (1)	
3. Oxygen	O ₂	Diatomic (2)	
4. Hydrogen	H ₂	Diatomic (2)	
5. Phosphorus	P ₄	Tetratomic (4)	
6. Sulphur	S ₈	Polyatomic (8)	
7. Ozone	O ₃	Triatomic (3)	

Chemical formulae

It is the symbolic representation of the composition of a compound.

Characteristics of chemical formulae

- The valencies or charges on ion must balance.
- When a compound is formed of metal and non-metal, symbol of metal comes first. *E.g.*, CaO, NaCl, CuO.
- When polyatomic ions are used, the ions are enclosed in brackets before writing the number to show the ratio. *E.g.*, Ca(OH)₂, (NH₄)₂ SO₄

Molecular Mass

It is the sum of atomic masses of all the atoms in a molecule of that substance.

E.g., Molecular mass of H₂O = 2 x Atomic mass of Hydrogen + 1 x Atomic mass of Oxygen

So, Molecular mass of H₂O = 2 x 1 + 1 x 16 = 18 u

Formula Unit Mass

It is the sum of atomic mass of ions and atoms present in formula for a compound.

E.g., In NaCl, Na = 23 a.m.u. , Cl = 35.5 a.m.u.

So, Formula unit mass = 1 x 23 + 1 x 35.5 = 58.5 u

Rule for writing chemical formulae

Rule-1 (i) We first write symbols of elements which form compound.

(ii) Below the symbol of each element, we should write their valency.

(iii) Now crisscross the valencies of combining atoms.

(iv) with first atom, we write the valency of second atom (as a subscript).

(v) with second atom, we write the valency of first atom (subscript).

Rule-2 When the valency is one, subscript is not written.

Rule-3 When there are multiple number of polyatomic ion, bracket must be used to separate the Polyatomic ion from subscript.

Examples:



[Take 2 common and divide the formula by 2]

(iii) For Hydrochloric acid (Hydrogen chloride)



(iv) For Carbon tetrachloride



(v) For Magnesium chloride



(vi) For aluminium oxide



(vii) For Calcium oxide



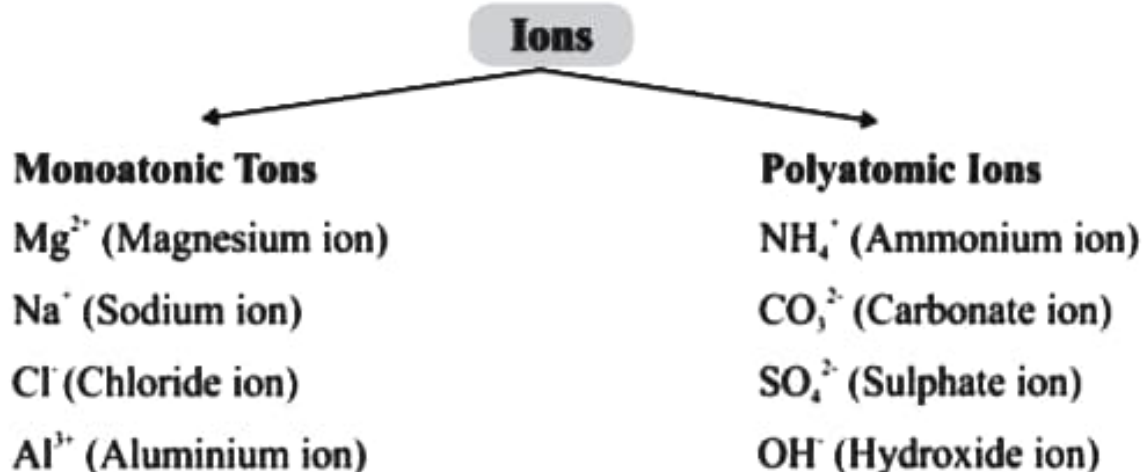
[Take 2 common and divide the formula by 2]

Ions

An ion may be defined as an atom or group of atoms having positive or negative charge.

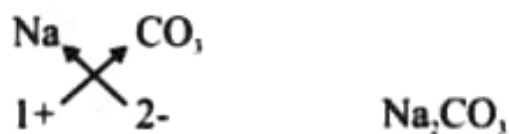
Some positively charged ions : Na^+ , K^+ , Ca^{2+} , Al^{3+}

Some negatively charged ions : Cl^- (chloride ion), S^{2-} (sulphide ion), OH^- (hydroxide ion), SO_4^{2-} (sulphate ion)



Chemical Formulae of Ionic Compounds (Polyatomic)

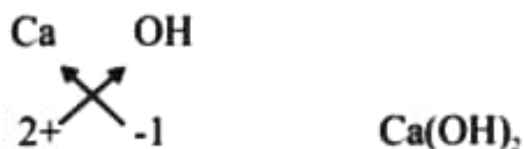
(i) Sodium carbonate



(ii) Aluminium sulphate



(iii) Calcium hydroxide



(iv) Ammonium sulphate



Molar Mass

The molar mass of a substance is the mass of 1 mole of that substance.

It is equal to the 6.022×10^{23} atoms of that element/substance.

Example :

- (a) Atomic mass of hydrogen (H) is 1 u. Its molar mass is 1 g/mol.
- (b) Atomic mass of nitrogen is 14 u. So, molar mass of nitrogen (N) is 14 g/mol.
- (c) Molar mass of S_8 = Mass of S $\times$ 8 = $32 \times 8 = 256$ g/mol
- (d) Molar mass of HCl = Mass of H + Mass of Cl
 $= 1 + 35.5 = 36.5$ g/mol

Mole concept

A group of 6.022×10^{23} Particles (atoms, molecules or ions) of a substance is called a mole of that substance.

$$1 \text{ mole of atoms} = 6.022 \times 10^{23} \text{ atoms}$$

$$1 \text{ mole of molecules} = 6.022 \times 10^{23} \text{ molecules}$$

Example, 1 mole of oxygen = 6.022×10^{23} oxygen atoms

6.022×10^{23} is Avogadro Number (L).

- 1 mole of atoms of an element has a mass equal to gram atomic mass of the element.

Important Formulae

$$(i) \text{ Number of moles } (n) = \frac{\text{Given mass}}{\text{Molar mass}} = \frac{m}{M}$$

$$(ii) \text{ Number of moles } (n) = \frac{\text{Given number of particles}}{\text{Avogadro's number}}$$

$$n = \frac{N}{N_0}$$

$$(iii) \frac{m}{M} = \frac{N}{N_0} \quad m = \frac{M \times N}{N_0}$$

$$(iv) \text{ Percentage of any atom in given compound} = \frac{\text{Mass of element} \times 100}{\text{Mass of compound}}$$

Example. Calculate no. of iron atoms in a piece of iron weighing 2.8 gm (At mass = 54 u).

Solution : 1 mole of iron = 56 gm (Gram atomic mass of iron)

1 mole of iron element contains 6.022×10^{23} atoms of iron.

So, 56 gm of iron = 6.022×10^{23} atoms

$$\begin{aligned} 2.8 \text{ gm of iron} &= \frac{6.022 \times 10^{23} \times 2.8}{56} \\ &= 3.011 \times 10^{22} \text{ atoms} \end{aligned}$$

Example. Mass of one molecule of a substance is 5.32×10^{-23} g. What is its molecular mass ?

Solution : Mass of 1 molecule of substance

$$= 5.32 \times 10^{-23} \text{ g}$$

Mass of 6.022×10^{23} molecules of substance

$$= 5.32 \times 10^{-23} \times 6.022 \times 10^{23}$$

$$= 32 \text{ g}$$

Example. Calculate the mass of 0.5 mole of N_2 gas.

Solution: 1 mole of N_2 = Gram molecular mass of N_2

Or 1 mole of N_2 = 28 gm

$$\begin{aligned} \therefore 0.5 \text{ mole of } N_2 \text{ gas} &= 0.5 \times 28 \\ &= 14 \text{ gm of } N_2 \end{aligned}$$

Example. Calculate the total number of O_2 molecules present in 8 gm of O_2 .

Solution: Gram molecular mass of O_2

$$= 6.022 \times 10^{23} O_2 \text{ molecules}$$

Or 32 gm of O_2 = $6.022 \times 10^{23} O_2$ molecules

$$\begin{aligned} \text{Or } 8 \text{ gm of } O_2 &= 6.022 \times 10^{23} \times 8/32 O_2 \text{ molecules} \\ &= 1.51 \times 10^{23} O_2 \text{ molecules} \end{aligned}$$

QUESTIONS

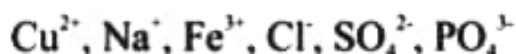
VERY SHORT ANSWER TYPE QUESTIONS

1. Name two laws of Chemical combination.
2. What is atomicity.
3. State law of conservation of mass.
4. State law of constant proportion.
5. Calculate molecular mass of CO_2 .
(At. mass of C = 12 u, At. Mass of O = 16u)
6. In what form do atoms of noble gases occur in nature.
7. Define molecular mass.
8. What do you understand by term 1 mole.
9. Write the chemical symbols of nitrogen gas and oxygen gas.
10. Name the elements by reading the given symbols.
Na, K, Ar, Ne, N, Mg, Al, Ca.

SHORT ANSWER TYPE QUESTIONS

1. Write the chemical formulae of-

(a) Calcium chloride	(e) Lead Nitrate
(b) Magnesium bicarbonate	(f) Calcium Phosphate
(c) Aluminum sulphate	(g) Iron (II) sulphide
(d) Sodium carbonate	(h) Mercury (I) chloride.
2. Write the molecular formulae of all the compounds that can be formed by the combination of following ions.



3. Write the cations (Positively ions) and anions (negatively charged ions)

Present (If any) in the following compounds.

- (a) NaCl (c) NH_4NO_3
(b) H_2 (d) $\text{Ca}(\text{HCO}_3)_2$

4. Give the formulae of the compounds formed from the following sets of elements

- (a) Calcium and fluorine (d) Sulphur and Oxygen
(b) Nitrogen and Hydrogen (e) Carbon and Oxygen
(c) Nitrogen and Oxygen (f) Carbon and Chlorine

5. Classify each of the following on the basis of their atomicity.

- (a) F_2 (b) NO_2 (c) CH_4 (d) P_4 (e) H_2O_2
(f) P_4O_{10} (g) O_3 (h) HCl (i) He (j) Ag

6. Calculate the number of moles of magnesium present in a magnesium ribbon weighing 12 gm. Molar atomic mass of Magnesium is 24 gm/mol.

7. write postulates of Dalton's atomic theory (atleast three).

8. what is the difference between the molecules of an element and the molecule of a compound? Give one example of each.

9. What is the difference between 2H and H_2 ? (atleast 2 dif.)

10. (a) what would be gm atomic mass of 5 moles of chlorine?

(b) Calculate the gm atomic mass of one atom of oxygen.

(gm at. mass of oxygen = 16 gm.)

LONG ANSWER TYPE QUESTIONS

1. Verify by calculating that 5 moles of CO_2 and 5 moles of H_2O do not have the same mass.

[Hint : molar mass of CO_2 = 44 g and molar mass of H_2O = 18 g]

2. If you take 5 moles of carbon atoms in a container and your friend take 5 moles of sodium atoms in another container of same weight.

[Hint : molar mass of carbon = 12 gm. molar mass of sodium = 23 gm]

- (a) Whose container will be heavier?
(b) Whose container has more number of atoms?
3. Which has more number of atoms?
100 gm of N_2 or 100 gm of Ammonia NH_3

$$\left[\text{Hint : No. of atoms} = \frac{\text{mass}}{\text{molar mass}} \times 6.022 \times 10^{23} \right]$$

4. Hydrogen and oxygen combine in the ratio of 1:8 by mass to form water, What mass of oxygen gas would be required to react completely with 3 gm of Hydrogen gas?
5. (a) Which postulate of Dalton's atomic theory is the result of the law of conservation of mass?
(b) Which postulate of Dalton's atomic theory can explain the law of constant

Objective Type Questions

1. Which of the following statements is not true about an atom?
- Atoms are not able to exist independently
 - Atoms are the basic units from which molecules and ions are formed
 - Atoms are always neutral in nature
 - Atoms aggregate in large numbers to form the matter that we can see, feel or touch
2. The chemical symbol for nitrogen gas is
- Ni
 - N_2
 - N^+
 - N

3. The Chemical symbol for sodium is
 - a. So
 - b. Sd
 - c. NA
 - d. Na
4. Which of the following correctly represents 360 g of water?
 - i. 2 moles of water.
 - ii. 20 moles of water
 - iii. 6.022×10^{23} molecules of water
 - iv. 1.2044×10^{25} molecules of water
 - a. i.
 - b. i.and iv
 - c. ii and iii
 - d. ii and iv
5. Give the formulae of the formed from the following sets of elements
 - a. Calcium and fluorine
 - b. Hydrogen and sulphur
 - c. Nitrogen and hydrogen
 - d. Carbon and chlorine
 - e. Sodium and oxygen
 - f. Carbon and oxygen
6. Write the molecular formulae for the following compounds
 - a. Copper (II) bromide.
 - b. Aluminium (III) nitrate.
 - c. Calcium (II) phoshate
 - d. Iron (III) sulphide
 - e. Mercury (II) chloride
 - f. Magnesium (II) chloride

7. Write the molecular formulae of the compounds that can be formed by the combination of following ions

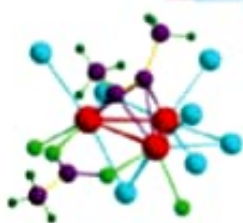
- a. Cu^{2+} and Cl^-
- b. Na^+ and NO_3^-
- c. Fe^{3+} and SO_4^{2-}
- d. Fe^{3+} and Cl^-

8. classify each of the following on the basis of their atomicity.

Elements	Atomicity
F_2	
NO_2	
N_2O	
P_4	
H_2O_2	
He	
Ag	
CH_4	
P_4O_{10}	

9. Fill in the blanks

- a. In a chemical reaction, the sum of the masses of the reactants and product remains unchanged. This is called
- b. A group of atoms carrying a fixed charge on them is called
- c. The formula unit mass of $\text{Ca}_3(\text{PO}_4)_2$ is
- d. Formula of sodium carbonate is and that of ammonium sulphate is



Chapter - 3

Atoms And Molecules

CONCEPT MAPPING

